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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert.**

NOTATION

In this study, the following notation is used:

- $\mathbb{R}^n$ denotes the n-dimensional Euclidean space.
- $\mathbb{R}^{n\times m}$ denotes the set of $n \times m$ real matrices.
- A vector and a matrix are denoted by $\mathbf{x} = [x_i] \in \mathbb{R}^n$ and $\mathbf{A} := [a_{ij}] \in \mathbb{R}^{n\times m}$, respectively.
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix, and $\mathbf{0}_{n\times m}$ denotes the $n \times m$ zero matrix.
- $\mathrm{vec}(\mathbf{A})$ denotes the vectorization of the matrix $\mathbf{A}$.
- A matrix $\mathbf{P} \succ 0$ ($\mathbf{P} \succeq 0$) is positive definite (positive semidefinite).
- $\lambda_{\min}(\mathbf{A})$ denotes the minimum eigenvalue of the matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$.
- $\otimes$ denotes the Kronecker product [**?**].

I. INTRODUCTION

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II. PROBLEM FORMULATION AND CONTROL OBJECTIVES

A. System Dynamics

In this study, the four-wheel-steering (4WS) vehicle is modeled as a simplified 2-DOF bicycle model. The following 2-DOF vehicle lateral dynamics are considered:

$$\dot{v}_y = \tfrac{1}{m}\left(F_{y,f} + F_{y,r}\right) - v_x\dot{\psi}, \quad \ddot{\psi} = \tfrac{l_f F_{y,f} - l_r F_{y,r}}{I_z}, \tag{1}$$

where v_x is the longitudinal velocity, v_y is the lateral velocity, and ψ is the yaw angle, m is the vehicle mass and I_z is the yaw moment of inertia, l_f and l_r are the distances from the center of gravity to the front and rear axles, respectively, and $F_{y,f}$ and $F_{y,r}$ are the front and rear lateral tire forces, respectively, and modelled as:

$$F_{y,f} = F_{\max}\tanh\left(\tfrac{C_f}{F_{\max}}\alpha_f\right), \tag{2}$$

$$F_{y,r} = F_{\max}\tanh\left(\tfrac{C_r}{F_{\max}}\alpha_r\right), \tag{3}$$

where C_f and C_r are the front and rear cornering stiffnesses, respectively, $F_{\max}$ is the saturation limit of the tire forces, and

α_f and α_r are the front and rear tire slip angles respectively and are given by:

$$\alpha_f = \theta_f - \tfrac{v_y + l_f\dot{\psi}}{v_x}, \qquad \alpha_r = \theta_r - \tfrac{v_y - l_r\dot{\psi}}{v_x}, \tag{4}$$

where θ_f and θ_r are the effective front and rear steering angles respectively. In order to capture the lag between the steering commands and the actual steering angles, the actuator dynamics are modeled as:

$$\dot{\theta}_f = \tfrac{\omega_f - \theta_f}{\tau}, \qquad \dot{\theta}_r = \tfrac{\omega_r - \theta_r}{\tau}, \tag{5}$$

where ω_f and ω_r are the front and rear steering commands respectively, and $\tau > 0$ is the actuator time constant.

The global position kinematics are given by (6):

$$\dot{X}_a = v_x\cos\psi - v_y\sin\psi, \qquad \dot{Y}_a = v_x\sin\psi + v_y\cos\psi, \tag{6}$$

where (X_a, Y_a) denotes the vehicle position in the global frame, and ψ denotes the yaw angle. The yaw rate $\dot{\psi}$ is obtained by integrating the yaw acceleration $\ddot{\psi}$ in (1).

B. Control Objectives

- end to end controller for path tracking

- we expect 1D CNN operation to extract underlying dynamics from stacked input(i.e. error derivatives, integrals, etc.) and provide informative features for steering control without an explicit information.

- we expect the controller adapts the vehicle paramters changes online(eg. tire characteristics,vehicle mass, etc.)

- we expect controller adapt its parameters online to handle the changing dynamics and maintain good tracking performance across different path segments (e.g., straight vs. curved) and varying conditions (e.g., different speeds, road friction, etc.).

III. System Details

A. CVL-CONAC Architecture

The network $\mathbf{\Phi}$ is designed as a hybrid architecture suitable for capturing spatio-temporal dynamics from historical data[??].

1) Convolutional Layers (CLs): The CVLs are represented recursively as:

$$\Phi^C_{j_c} = \begin{cases} O(\phi^C_{j_c}(\Phi^C_{j_c-1}), \Omega_{j_c}, B_{j_c}), & j_c \in [1, \ldots, k_c] \\ O(\phi^C_0(X), \Omega_0, B_0), & j_c = 0 \end{cases}, \tag{7}$$

where $X \in \mathbb{R}^{n_0 \times m_0}$ denotes the network input matrix, $O : \mathbb{R}^{n_{j_c} \times m_{j_c}} \times \mathbb{R}^{p_{j_c} \times m_{j_c} \times q_{j_c}} \times \mathbb{R}^{q_{j_c}} \to \mathbb{R}^{n_{j_c+1} \times m_{j_c+1}}$ denotes the CNN operator (see Appendix A), and $\phi^C_{j_c} : \mathbb{R}^{n_{j_c} \times m_{j_c}} \to \mathbb{R}^{n_{j_c} \times m_{j_c}}$ denotes the matrix activation function (i.e., $\phi^C(\Phi^C_{j_c-1})_{(i,j)} = \sigma_{j_c}(\Phi^C_{j_c-1})_{(i,j)}$) for some activation function $\sigma_{j_c} : \mathbb{R} \to \mathbb{R}$. The first activation function ϕ^C_0 should be a bounded nonlinear function to ensure that the input to the first CNN operator is bounded. In this study, $\alpha_1 \tanh(\cdot)$ is selected with $\alpha_1 \in \mathbb{R}_{>0}$. The filter set Ω_{j_c} contains q_{j_c} filters $W^{(i)}_{j_c} \in \mathbb{R}^{p_{j_c} \times m_{j_c}}$, $\forall i \in [1, \ldots, q_{j_c}]$, and is represented as $\Omega_{j_c} = \{W^{(1)}_{j_c}, \ldots, W^{(q_{j_c})}_{j_c}\}$, where superscript i denotes the filter index. The bias vector B_{j_c} consists of q_{j_c} biases.

The weights for the j_c-th CVL layer are collectively represented as $\tilde{H}_{Cj_c} \triangleq [\mathrm{vec}(\Omega_{j_c})^T, B^T_{j_c}]^T$.

2) Fully-Connected Layers (FCLs): The output matrix of the CVLs is input to the concatenate layer $C(\Phi^C_{k_c}) = [\mathrm{vec}(\Phi^{C^\top}_{k_c})^\top, 1]^\top$ before the input layer of FCLs for compatibility between the CVLs and FCLs. The FCLs are represented recursively as:

$$\Phi^F_{j_f} = \begin{cases} V^\top_{j_f}\phi^F_{j_f}(\Phi^F_{j_f-1}), & j_f \in [1, \ldots, k_f] \\ V^\top_0 C(\Phi^C_{k_c}), & j_f = 0 \end{cases}, \tag{8}$$

where $V_{j_f} \in \mathbb{R}^{l_{j_f}+1 \times l_{j_f+1}}$ denotes the weightmatrix, $\phi^F_{j_f} : \mathbb{R}^{l_{j_f}} \to \mathbb{R}^{l_{j_f}+1}$ denotes the vector activation function defined by $\phi^F_{j_f}(\Phi^F_{j_f-1}) \triangleq [\sigma_{j_f}(\Phi^F_{j_f-1})^\top, 1]^\top$ for $j_f \in [1, \ldots, k_f - 1]$, for some nonlinear activation functions $\sigma_{j_f} : \mathbb{R} \to \mathbb{R}$ such as $\tanh(\cdot)$. Note that $C(\Phi^C_{k_c})$ and $\Phi^F_{j_f}$ are augmented by 1 to consider the bias of the preceding input layer as a weight in the weight matrix. The output of the FCLs is the final output of the CNN architecture and is represented as $\mathbf{\Phi} \equiv \Phi^F_{k_f}$.

The FCL weights for the j_f-th layer are $\tilde{H}_{Fj_f} \triangleq \mathrm{vec}(V_{j_f})$.

IV. Adaptation Laws

A. Optimization Problem

V. Stability Analysis

VI. NUMERICAL VALIDATION

A. Validation Setup

In this section, the proposed controller is validated through numerical simulation using the nonlinear 4WS vehicle model described in Section II. The desired trajectory is selected as a closed stadium-shaped path composed of two straight segments and two semicircular arcs. Let L and R denote the straight length and turning radius, respectively. Then, the lower and upper straight segments are defined by

$$\mathcal{P}_1 = \{(x, y) \,|\, 0 \le x \le L,\ y = 0\}, \tag{9}$$

$$\mathcal{P}_2 = \{(x, y) \,|\, 0 \le x \le L,\ y = 2R\}, \tag{10}$$

The left and right turning sections are semicircles centered at $C_{\mathrm{L}} = (0, R)$ and $C_{\mathrm{R}} = (L, R)$, respectively. This path contains both zero-curvature and constant-curvature segments and is therefore suitable for evaluating straight-line regulation and turning performance within a single experiment. Figure 1 illustrates the stadium path together with the vehicle pose, body-fixed frame, and tracking error used in the validation setup.

To generate the path-following reference at each simulation step, the current vehicle position (X_a, Y_a) is projected onto the corresponding path segment. For straight segments, the desired heading is:

$$\psi_{\mathrm{des}} = \begin{cases} 0, & \text{lower straight}, \\ \pi, & \text{upper straight}, \end{cases} \tag{11}$$

For the arc segments, the projected point is computed by

$$X^\star = x_c + R\cos\theta, \qquad Y^\star = y_c + R\sin\theta, \tag{12}$$

with

$$\theta = \mathrm{atan2}(Y_a - y_c, X_a - x_c), \tag{13}$$

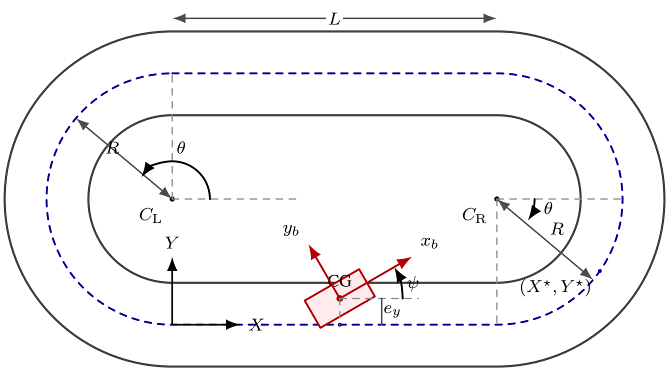

Fig. 1. Stadium-track geometry and vehicle pose used in the validation setup.

The desired heading is then taken as the tangent direction of the arc,

$$\psi_{\mathrm{des}} = \theta + \frac{\pi}{2}, \tag{14}$$

The tracking errors are then defined by the lateral displacement from the projected point and the heading error.

B. Validation Results

The validation results are presented in terms of tracking errors, steering commands, and the global vehicle trajectory. In particular, the lateral and heading errors are used to assess whether the proposed 1D CNN–MLP controller can regulate the vehicle toward the desired path while adapting its parameters online. The front and rear steering-rate commands illustrate the control effort generated by the end-to-end policy, and the global XY trajectory provides a direct geometric comparison between the desired path and the actual motion.

In the final manuscript, this subsection should be organized around the simulation plots produced by the current MATLAB implementation. A compact presentation consistent with the reference paper would first discuss the validation conditions, then summarize the tracking behavior in the time domain, and finally comment on the overall path-following accuracy and adaptation behavior. Root-mean-square tracking errors and selected controller settings can also be reported in a table to make the comparison more concise.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks . . .". Instead, try "R. B. G. thanks. . .". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

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